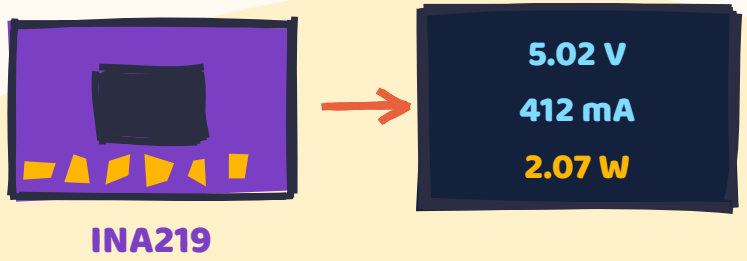
Young Innovators · Class 27

Energy Detective: How Much Power Do Gadgets Use?

Measure it. Price it. Cut it.

Watts vs units | INA219 | Savings



Did you know?
A phone charger left plugged in with no phone still uses a little power, all day, every day. Detectives call these 'vampire loads'.

LEARN

Let's think...

Which costs more: a fan or a phone charger?

The bill mystery

Every month the electricity bill arrives with a number of **units**. But which gadget used them? The AC? The fridge? The charger left in the wall? Today you become energy detectives and find out with real measurements.

5.02 V

412 mA

2.07 W

The AC, obviously!

The charger?

Measure it!

Today: you measure real gadgets with a **power sensor**, calculate their monthly cost, and find the biggest wasters.

Learning objectives

By the end of this class I can...

1	**Power vs energy:** explain the difference between **power (W)** and **energy (Wh, kWh = 1 unit)**	**Proof:** I can read the units on an electricity bill
2	**Measure it:** measure **voltage, current and power** of real gadgets with an **INA219** sensor	**Proof:** my Serial Monitor shows V, mA and mW for 3 gadgets
3	**Save energy:** calculate **energy and cost** per month and recommend a saving backed by data	**Proof:** our group's poster shows the biggest energy wasters

Objective 1 · Power vs energy · video

Video time!



What is a kilowatt hour? Understanding home energy use

Enphase Energy · about 3 minutes · youtu.be/zRYESRObKqA

Watch for...

1. What is a **kilowatt-hour**?
2. How do you work out how much energy a gadget uses?
3. Which home gadgets use the **most** energy?

Scan to watch

After the video: Guess: which gadget in your home uses the most units each month? We will check today.

LEARN

Objective 1 · Power vs energy

Power is a rate; energy is the total

The idea

Power (watts, W): how fast energy is used right now. **P = V × I** (volts × amps).

Energy (watt-hours, Wh): power × time. A 100 W bulb for 10 hours = **1000 Wh = 1 kWh**.

On your bill, **1 unit = 1 kWh**. Cost = units × price per unit (check your own bill; in Pakistan often several tens of rupees per unit).

Example: speed and distance



100 W

× 10 h = **1 kWh**

= 1 unit

Speed (km/h) is how fast you go. Distance is speed × time. Power is like speed; energy is like distance. A slow car driven all day can travel further than a fast car driven for a minute.

Objective 2 · Measure it

How the INA219 measures

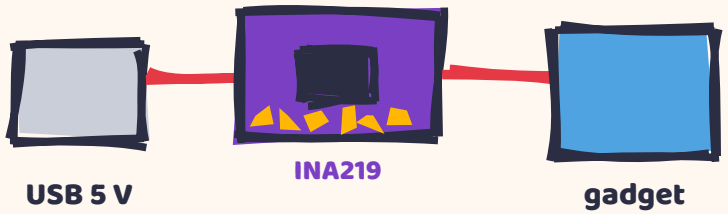
The idea

The INA219 sits **in series** on the + wire, so all the gadget's current flows through a tiny **0.1 Ω shunt** resistor.

It measures the tiny voltage across the shunt → **current** (Ohm's law, I = V ÷ R), and the voltage to ground → **bus voltage**. Then **P = V × I**.

It sends the numbers to Arduino over **I²C** (SDA, SCL). Limits: up to **26 V** and about **3.2 A**.

Example: a water meter



A water meter sits in the pipe and all the water flows through it, so it can count every litre. The shunt is a meter in the wire.

Objective 3 · Save energy

From measurement to money

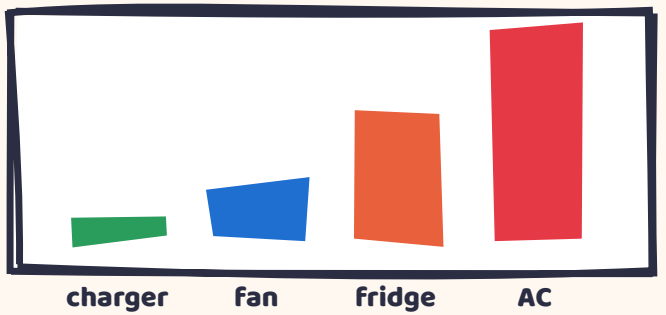
The idea

Measure the power **P** (W). Estimate **hours per day** it's used.

kWh per month = P × hours × 30 ÷ 1000
Cost = kWh × price per unit

Example: 2 W charger left in 24 h/day → 2 × 24 × 30 ÷ 1000 = **1.44 units/month**. Small... but 200 million chargers?
A 1500 W AC for 8 h/day → **360 units**!

Example: pocket money leaks



Rs 10 a day doesn't feel like much. Over a month it's Rs 300. Energy leaks work the same: small power, long time, big total.

Real world

Energy detectives at work

Smart meters
Show your home's power use live

Solar homes
Size panels and batteries from kWh data

Load-shedding
Grid operators balance supply and demand

Energy labels
Star ratings on ACs and fridges in shops

? If every home in your city saved 1 unit a day, what would that mean for load-shedding?

Watt it costs
Energy saved at home is energy the power station never needs to make, and fuel it never needs to burn.

Quick quiz

Energy maths

Question	Answer
1. 5 V × 0.4 A = ? W	
2. 60 W fan for 5 hours = ? Wh	
3. How many Wh in 1 unit?	
4. 10 W router on 24 h for 30 days = ? units	
5. Why does the INA219 go in series?	

Answers

Cover! 1: 2 W · 2: 300 Wh · 3: 1000 · 4: 7.2 · 5: all the current must flow through it to be measured

Part 2 · Make it

Now let's put it to work!

You know the ideas. Time to play, build and test. Use these slides while you work:

1. Power auction
2. Our detective kit (per group)
3. Read V, I, P and count energy
4. Investigate 3 gadgets
5. Price it and present
6. Energy log
7. Energy auditor challenges

MAKE

Class game

Power auction

How to play

1. Each team gets 10 'unit' tokens for one day at home.
2. Teacher shows gadget cards with their power: AC 1500 W, fan 60 W, LED 10 W, TV 100 W, iron 1000 W, charger 5 W…
3. Teams plan a summer day: which gadgets for how many hours? Pay 1 token per kWh.
4. Who stays cool and comfortable within budget?

Then discuss

Which single gadget ate most of your budget? What swaps saved the most?



100 W

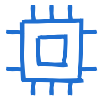
Big idea

High-power gadgets for many hours dominate the bill.

MAKE

Build it

Our detective kit (per group)



Arduino + INA219
Measures V, I, P



USB cable 'cut' adapter
Teacher-made, + wire goes through INA219



USB gadgets
LED lamp, USB fan, LED strip, phone charging



Kill-A-Watt plug meter
Teacher demo for mains gadgets



Energy survey sheet
Home gadgets and hours



Serial Plotter
Watch power change live

Wiring: INA219 VCC → 5V, GND → GND, SDA → A4, SCL → A5. Gadget + wire: VIN+ in, VIN− out.

Mains = NO
We only measure low-voltage USB gadgets ourselves. Mains (230 V) gadgets are measured by the teacher with a plug-in meter.

MAKE

Objective 2 · Measure it · Build step 1 of 3

Read V, I, P and count energy

```ino
#include <Wire.h>
#include <Adafruit_INA219.h>
Adafruit_INA219 ina;

float mWh = 0;                      // energy counter
unsigned long last = 0;

void setup() {
  Serial.begin(9600);
  if (!ina.begin()) { Serial.println("INA219 not found"); while (1); }
  last = millis();
}

void loop() {
  float V  = ina.getBusVoltage_V();
  float mA = ina.getCurrent_mA();
  float mW = ina.getPower_mW();

  unsigned long now = millis();
  mWh += mW * (now - last) / 3600000.0;   // power × time
  last = now;

  Serial.print(V, 2);  Serial.print(" V  ");
  Serial.print(mA, 0); Serial.print(" mA  ");
  Serial.print(mW, 0); Serial.print(" mW  ");
  Serial.print(mWh, 3); Serial.println(" mWh");
  delay(500);
}
```

Library

Install 'Adafruit INA219' from Library Manager.

Energy

Every half second we add power × time. That's how real smart meters count.

3600000

Milliseconds in one hour: turns mW × ms into mWh.

MAKE

Objective 2 · Build step 2 of 3

Investigate 3 gadgets

a Measure each USB gadget for 1 minute: LED lamp, USB fan (low and high), LED strip (one colour vs white).

b Record average **V, mA, mW** in your table.

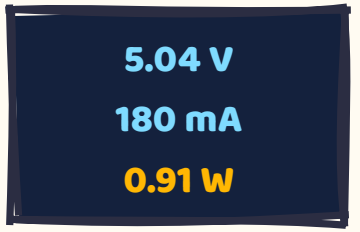
c Phone charging: measure at the start, and again when the battery is nearly full. What changes?

d Measure the charger with **no phone** connected (on the USB side).

e Teacher demo: plug meter on the class fan, kettle and projector.



INA219



Check: Your table has V, mA and mW for at least 3 gadgets.

Never cut or open a mains cable or plug. Only the teacher's prepared USB adapter.

MAKE

Objective 3 · Build step 3 of 3

Price it and present

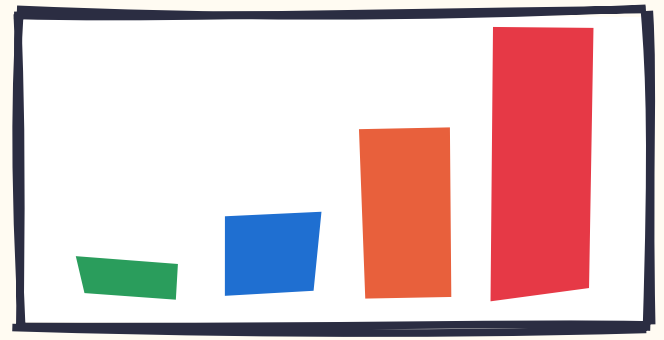
a For each gadget (and the teacher's mains ones), estimate **hours per day** at home.

b Calculate units per month = W × hours × 30 ÷ 1000, and cost = units × price.

c Make a bar chart: monthly units per gadget.

d Find the **biggest waster** and the **easiest saving**.

e Write one recommendation with numbers: 'Switching X saves Y units = Rs Z per month.'



Check: Your poster shows the biggest waster and a saving in rupees.

MAKE

Test like an engineer

Energy log

Gadget	Power	Unit
LED lamp		mW
USB fan low / high		mW
LED strip white vs one colour		mW
Phone charging start / end		mW
Charger, no phone		mW

Debug checklist

'INA219 not found'? Check SDA/SCL and that VCC is 5 V.
Current reads 0? The gadget's + wire must go THROUGH VIN+ and VIN–.
Negative current? VIN+ and VIN– swapped.

Real detectives repeat measurements. Take 3 readings and average them.

MAKE

Level up!

Energy auditor challenges

Choose your challenge



OLED meter: show V, mA and mWh on an OLED: a portable power meter!



Solar: measure a small solar panel's power at different angles.



Logger: log power every second to a CSV and plot a charging curve in Python.



School audit: survey every gadget in our classroom and estimate yearly units.

Think like an engineer

Why does the phone draw less power when its battery is nearly full?

Boss level

Make a 'vampire hunter': an alarm that beeps if a charger draws power with nothing attached.

Objective check

Can I really do it? Prove it.

01	What's the difference between watts and units (kWh)?	○ ○ ○
02	Show your measurements. Which USB gadget used the most power?	○ ○ ○
03	Present your biggest waster and your saving in rupees.	○ ○ ○

Rate yourself

Colour one circle each. Red: not yet. Amber: with help. Green: I can explain it to a friend.

Teacher: Energy pledge: each student picks one real saving to try at home this week, with the number of units it saves.

Show what you know

What did we learn?

- [x] Power (W) = V × I, right now
- [x] Energy (Wh) = power × time; 1 unit = 1 kWh
- [x] INA219 goes in series and measures V, I, P
- [x] Small power for a long time adds up
- [x] Data makes saving energy convincing

At home

Read your home's electricity bill: how many units last month?

Exit ticket: a 10 W LED for 100 hours uses how many units?

5.02 V
412 mA
2.07 W

Next class: Python Control Panel for Arduino!